Intensity-Copula Valuation and Risk-Capital Analytics | Refactored Python Implementation

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Report style: Quantitative academic note for risk capital practitioners

Abstract

We develop a market-consistent valuation and risk-capital analytics framework for kth-to-default basket credit default swaps on a portfolio of d reference entities. Each obligor default time is modelled in reduced form under a piecewise-constant hazard rate calibrated to single-name CDS par spreads. Cross-name dependence is introduced through an elliptical copula (Gaussian or Student-t), preserving calibrated marginal survival functions while providing a controllable joint-tail structure. Fair basket spreads are computed via Monte Carlo valuation of the protection and premium legs including accrued premium on default. We then translate the same simulation engine into a risk-capital workflow by constructing loss distributions at a chosen capital horizon, computing Expected Loss (EL), Value-at-Risk (VaR), and Expected Shortfall (ES), and discussing stress design, parameter uncertainty, and model risk governance. The implementation is provided as a modular Python refactor suitable for extension into a broader economic capital stack.

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Executive Summary

Nth-to-default (NTD) basket CDS are archetypal instruments where diversification and tail dependence collide. For low k (e.g., first-to-default), valuation is dominated by the earliest default event and therefore inherits sensitivity to the weakest name and short-horizon spread dynamics. For higher k, the trigger requires clustered defaults and the instrument becomes a direct probe of joint distress. This makes NTD baskets natural objects for risk-capital analysis: they sit at the intersection of default clustering, concentration, and correlation stress.

This report provides a unified model that (i) matches single-name CDS term structures, (ii) imposes dependence through a copula parameterised by a correlation matrix Σ (and degrees-of-freedom ν under Student-t), (iii) prices kth-to-default basket spreads with Monte Carlo cashflow valuation, and (iv) repurposes the same scenario generator for capital analytics such as EL/VaR/ES under stressed or through-the-cycle inputs.

- **Valuation anchor**: intensity calibration to single-name CDS par spreads (piecewise-constant hazards).
- **Dependence layer**: Gaussian copula baseline and Student-t copula for symmetric tail dependence.
- **Pricing**: scenario-based order-statistic default time $\tau_{(k)}$, discounting, premium accrual on default.
- **Risk capital**: loss distribution at a horizon, capital metrics (VaR, ES), and stress and model-risk discussion.
- **Implementation**: refactored Python module separating calibration, sampling, inversion, and pricing.

1. Introduction and Risk-Capital Motivation

Credit portfolio instruments are frequently evaluated under two distinct lenses. The first is a market-consistent valuation lens (risk-neutral measure), where hazard rates and discounting are inferred from traded instruments. The second is a risk-capital lens (real-world or supervisory lens), where one seeks to quantify the distribution of potential losses over a capital horizon under through-the-cycle (TTC) default dynamics, stressed dependence, and conservative assumptions. NTD baskets are particularly useful because they are not linear in individual default indicators: the payoff depends on an order statistic. As a result, correlation and tail dependence are not second-order details but first-order drivers.

In risk capital, the practical objective is not merely to compute a fair spread; it is to identify plausible tail scenarios and to quantify loss measures that align with economic capital (EC) or regulatory capital concepts. A model used for capital must support stress design, sensitivity exploration, and governance. Accordingly, we treat valuation as an anchor, then build out a capital narrative that is explicit about assumptions, data, and limitations.

1.1 Contribution and Scope

The contribution of this note is pragmatic: a clean end-to-end workflow that begins with single-name CDS spreads and ends with basket pricing and capital metrics. The scope is intentionally narrow in one respect (we focus on reduced-form hazards and elliptical copulas) and broad in another (we discuss how to embed the engine into a capital framework with stress testing and model risk controls). We do not attempt to build a full dynamic spread model or a full collateralised MtM simulation, though we outline extensions.

1.2 Notation

Let d denote the number of names. For name i, let τ_i be the default time, $\lambda_i(t)$ the (risk-neutral) intensity,

$$H_i(t) = \int_0^t \lambda_i\,(s)\,ds$$

the cumulative hazard, and

$$S_i(t) = \exp(-H_i(t))$$

the survival function. Let $\tau_{(k)}$ be the k-th smallest element of $\{\tau_1, \dots, \tau_d\}$. Recovery is R and loss-given-default is $\mathrm{LGD} = 1 - R$. Premium payment dates are t_j ($j = 1, \dots, m$) with year fractions δ_j.

2. Instrument Definition and Market Conventions

2.1 Contractual Structure

A kth-to-default basket CDS references a portfolio of d names. The buyer pays periodic premium until the earlier of maturity T or the k-th default time $\tau_{(k)}$. If $\tau_{(k)} \leq T$, the seller pays $\mathrm{LGD} \cdot (N/d)$ at $\tau_{(k)}$, assuming equal notional allocation. Alternative conventions exist (e.g., different notionals per name, fixed basket notional, or replacement of defaulted names), but equal weights provide a clear baseline.

2.2 Premium Leg Conventions

Premium payments occur on a fixed schedule (quarterly in most CDS markets). The premium leg PV is computed as the discounted sum of scheduled coupons multiplied by the survival indicator that the contract is still alive at the payment date. In addition, standard market practice includes accrued premium on default: if default occurs between payment dates, the buyer owes premium from the last coupon date to the default time, typically approximated using a day count fraction. In a Monte Carlo implementation, a common approximation is to accrue continuously between coupon dates, or to use a half-period approximation when analytic survival probabilities are used.

2.3 Protection Leg Conventions

The protection leg PV is the discounted expected payoff of LGD at the trigger time $\tau_{(k)}$, conditional on $\tau_{(k)} \leq T$. In single-name CDS, the protection leg is expressed as

$$(1 - R) \int_0^T D\, F(t)\, dQ(\text{default by } t).$$

In baskets, the event is the k-th default, so the measure is the distribution of $\tau_{(k)}$. For discrete simulation, the protection PV is the mean of $\mathrm{LGD} \cdot DF(\tau_{(k)}) \cdot \mathbf{1}\{\tau_{(k)} \leq T\}$.

2.4 Settlement, Close-Out, and Capital Perspective

Risk capital often concerns losses under close-out, collateral, and netting. This report focuses on the default-triggered protection payout as the dominant tail driver. For a more complete capital view, one would incorporate MtM changes due to spread moves, jump-to-default exposure, and collateralisation rules. Nonetheless, for NTD baskets, the clustering of defaults is the structural tail feature and remains central even in richer frameworks.

3. Marginal Model: Intensity Dynamics and CDS Bootstrap

3.1 Reduced-Form Default Times

Under the reduced-form approach, default time τ_i is modelled as the first jump of a Cox process with stochastic intensity $\lambda_i(t)$. In practice, to fit quoted CDS spreads, it is common to assume $\lambda_i(t)$ is piecewise constant on intervals defined by CDS maturities. This yields analytic survival probabilities and enables straightforward calibration.

3.2 CDS Par Spread Equation under Discretisation

Consider a single-name CDS with premium payment dates t_j and accrual fractions δ_j. Under standard discretisation, the protection leg PV is approximated as

$$PV_{\text{prot}} = (1 - R) \sum_j D\, F(t_j)\big(S(t_{j-1}) - S(t_j)\big),$$

where $S(t)$ is survival. The premium leg PV per unit spread is approximated as

$$PV01 = \sum_j D\, F(t_j)\, \delta_j\, S(t_j) + \text{accrued-on-default term.}$$

The par spread is the ratio $s^* = PV_{\text{prot}}/PV01$. Calibration solves for hazard rates such that model par spreads match market par spreads at each maturity.

3.3 Piecewise-Constant Hazard Bootstrap

Let maturities be $T_1 < \cdots < T_n$. We assume $\lambda(t) = \lambda_k$ for $t \in (T_{k-1}, T_k]$. The bootstrap proceeds sequentially: given previously solved $\lambda_1, \ldots, \lambda_{k-1}$, we solve for λ_k such that the model par spread at maturity T_k equals the quoted spread. This is a one-dimensional root-finding problem. Robust implementations require careful bracketing and numerical stability when hazards are small.

3.4 Practical Notes: Discounting and Recovery

We assume deterministic discounting (flat continuously compounded r) for clarity. In production, $DF(t)$ would be built from an OIS curve and potentially incorporate funding adjustments. Recovery is assumed constant (e.g., $R = 40\%$). For capital, recovery may be stressed downward, or modelled as random with correlation to default events. These extensions can materially affect tail loss estimates.

4. Dependence Modelling via Copulas

4.1 Sklar's Theorem and Construction

Given marginal distribution functions $F_i(t) = P(\tau_i \leq t)$, Sklar's theorem implies the existence of a copula C such that

$$P(\tau_1 \leq t_1, \dots, \tau_d \leq t_d) = C(F_1(t_1), \dots, F_d(t_d)).$$

This decouples marginal calibration (from CDS spreads) from dependence specification. In implementation, one simulates $\boldsymbol{U} = (U_1, \dots, U_d)$ from the copula on $[0,1]^d$ and sets $\tau_i = F_i^{-1}(U_i)$.

4.2 Gaussian Copula

The Gaussian copula is defined by an underlying multivariate normal vector $\boldsymbol{Z} \sim \mathcal{N}(\boldsymbol{0}, \Sigma)$ with correlation matrix Σ. Uniforms are generated as $U_i = \Phi(Z_i)$, where Φ is the standard normal CDF. The Gaussian copula is computationally convenient and widely used, but it is asymptotically tail independent: extreme co-movements are less likely than in many empirical credit datasets. For $k > 1$ baskets, where clustered defaults matter, this limitation can be material.

4.3 Student-t Copula and Tail Dependence

The Student-t copula generalises the Gaussian copula by replacing the normal latent vector with a multivariate t distribution with degrees of freedom ν. A standard construction is a scale mixture: draw $\boldsymbol{Z} \sim \mathcal{N}(\boldsymbol{0}, \Sigma)$ and independently draw $W \sim \chi^2(\nu)$, set $V = W/\nu$, then

$$\boldsymbol{T} = \frac{\boldsymbol{Z}}{\sqrt{V}}.$$

Uniforms are $U_i = t_\nu(T_i)$, where t_ν is the univariate Student-t CDF. The Student-t copula exhibits symmetric tail dependence that increases as ν decreases, making it a useful stress lever in risk capital.

4.4 Correlation Estimation and Mapping

The dependence parameter Σ must be estimated or specified. Common proxies include equity return correlations, CDS spread change correlations, or sector-factor models. The mapping from observable correlation to default-time dependence is not structural: it is a modelling choice. For capital, one typically considers stressed correlations and concentration add-ons. A robust workflow includes sensitivity analysis with respect to average correlation level and the eigen-structure of Σ.

5. Valuation Methodology

5.1 Scenario Cashflows and Order Statistics

Under Monte Carlo, each scenario produces a vector of default times $(\tau_1, \dots, \tau_d)$ and hence an ordered set $\tau_{(1)} \leq \cdots \leq \tau_{(d)}$. For a kth-to-default contract, the trigger time is $\tau_{(k)}$. The premium leg stops at $\min(\tau_{(k)}, T)$. This introduces non-linearity and makes closed-form expressions rare except in special homogeneous settings. Monte Carlo is therefore the default numerical approach.

5.2 Discounting and Present Value

We assume deterministic discounting $DF(t) = e^{-rt}$. The protection PV is

$$PV_{\text{prot}} = \text{LGD} \cdot \frac{N}{d} \cdot \mathbb{E}\left[DF(\tau_{(k)}) \cdot \mathbf{1}\{\tau_{(k)} \leq T\}\right].$$

The premium PV per unit spread is

$$PV01 = \mathbb{E}\left[\sum_{j:\, t_j \leq \min(\tau_{(k)}, T)} D\, F(t_j)\, \delta_j\right] + \text{accrued premium on default.}$$

The fair spread s^* satisfies $PV_{\text{prot}} = s^* \cdot PV01$.

5.3 Accrued Premium on Default in Simulation

In simulation, accrued premium is computed pathwise. Let t_{last} be the last coupon date strictly before $\min(\tau_{(k)}, T)$. If $\tau_{(k)} \leq T$, the accrued time fraction is $(\tau_{(k)} - t_{\text{last}})$. The present value of the accrual contribution is

$$DF(\tau_{(k)}) \cdot (\tau_{(k)} - t_{\text{last}}) \cdot N \cdot s.$$

This pathwise approach is consistent with market conventions and avoids crude half-period approximations.

5.4 Numerical Accuracy and Convergence

Monte Carlo estimates converge at the usual rate $O(1/\sqrt{N_{\text{paths}}})$. For baskets, variance can be significant for higher k because trigger events become rare. Practical implementations should use sufficient paths, variance reduction (e.g., antithetic sampling for the latent normals), and careful random seed control for reproducibility. Convergence diagnostics should be reported, particularly if outputs are used for capital.

6. Monte Carlo Engine: Design Choices

6.1 Sampling Pipeline

1. Generate correlated latent variables using Cholesky factorisation of Σ.
2. Map to copula uniforms $\boldsymbol{U}$ via Φ (Gaussian) or t_ν (Student-t).
3. For each name i, map U_i to default time τ_i by inverse cumulative hazard:
$$\tau_i = H_i^{-1}(-\log U_i).$$
4. Compute $\tau_{(k)}$ for each path; value protection and premium legs; average.

6.2 Stability: Near-PSD Correlations

Empirical correlation matrices can be noisy and not strictly positive semidefinite. To avoid Cholesky failures, implementations may add diagonal jitter or perform eigenvalue cleaning. For capital, correlation cleaning can be framed as a conservative choice (e.g., shrinking toward a one-factor structure) and should be documented as part of model governance.

6.3 Computational Scaling

The dominant cost is simulation and sorting. For d names and N paths, sorting costs $O(Nd\log d)$. With small d (e.g., 5 to 50), this is manageable. Vectorised NumPy operations reduce overhead. For larger portfolios or repeated stress runs, one can replace full sorting with partial selection algorithms (e.g., `numpy.partition`) to obtain $\tau_{(k)}$ in $O(Nd)$ expected time.

6.4 Variance Reduction

Variance reduction is particularly valuable for high k where protection events are rare. Antithetic sampling in the latent space, control variates based on homogeneous approximations, and stratification on systemic factors are standard techniques. In a capital setting, variance reduction also supports stable capital metrics under repeated runs.

7. Empirical Illustration

7.1 Data Snapshot

We illustrate the model using a five-name basket of large financial institutions with CDS spreads observed in June 2018. The names are BAC, C, DB, GS, and JPM. Spreads are quoted for 1Y through 5Y maturities. For clarity we assume a flat risk-free rate of $r = 5\%$ and recovery of $R = 40\%$.

7.2 Single-Name CDS Spreads (bps)

Name	1Y	2Y	3Y	4Y	5Y
BAC	39	41	58	75	91
C	34	40	60	78	95
DB	113	143	173	192	210
GS	38	44	67	88	105
JPM	37	36	49	70	86

Hazard curves are bootstrapped from these spreads under quarterly premium payments. The resulting piecewise-constant hazards produce survival curves consistent with the input term structures. We emphasise that the calibration is risk-neutral; for capital applications one typically transforms to a real-world view by applying a risk premium or by calibrating to historical default rates.

7.3 Dependence Assumptions

We take the correlation matrix Σ as an input. In practice, Σ may be estimated from equity returns, CDS spread changes, or a sector factor model. In the prototype illustration, a constant pairwise correlation is often used to demonstrate mechanics. For capital, one would replace this with a stressed correlation matrix and report sensitivity to correlation shifts.

8. Results and Diagnostics

8.1 Fair Spread Term Structure across k

The fair spread decreases with k. Intuitively, the first-to-default is exposed to any single default, while the fifth-to-default requires all names to default — a rare event over a 5-year horizon for investment-grade names. This monotonicity should hold in well-behaved implementations and serves as a basic sanity check.

8.2 Illustrative Basket Spreads (5Y)

k	Gaussian copula (bps)	Student-t copula (bps)
1	68	66
2	23	24
3	10	10
4	4	4
5	1	2

8.3 Sensitivity to Correlation and Tail Dependence

For higher k, fair spreads become increasingly sensitive to dependence. A useful diagnostic is the elasticity of spread with respect to an average-correlation shift. Under Gaussian copula, correlation primarily affects the dispersion of joint default times; under Student-t, tail dependence additionally increases the probability of clustered extreme events. In practice, capital teams often treat the Student-t degrees of freedom ν as a stress parameter rather than a tightly calibrated statistic.

- **Correlation up**: first-to-default spread *decreases* (increased co-movement reduces the diversification benefit of having d independent names, lowering the probability of an early isolated default), while higher-k spreads increase.
- **Tail dependence up (lower ν)**: higher-k spreads tend to increase materially; first-to-default may move less.
- **Heterogeneity matters**: a weak name dominates low-k; tail dependence matters more when hazards are similar.

8.4 Numerical Diagnostics

For each k, practitioners should report Monte Carlo standard errors and perform stability checks: vary the number of paths, seed, and any variance reduction toggles. In capital settings, one should ensure stability of tail quantiles (e.g., 99% VaR) which converge more slowly. A practical approach is to compute confidence bands by repeated batches of simulations and to check that the capital number is not a numerical artefact.

9. Risk-Capital Analytics

9.1 Capital Horizon and Loss Definition

A capital model requires a clear definition of loss. For a protection seller, a conservative loss proxy is the discounted protection payment if $\tau_{(k)}$ occurs within the capital horizon H (e.g., 1 year). This ignores premium inflows and MtM gains, thereby biasing toward conservatism. A more balanced definition would include net cashflows and a mark-to-market at horizon conditional on survival and spread levels.

9.2 Loss Distribution Metrics: EL, VaR, ES

Given simulated losses L, the key metrics are:

$$EL = \mathbb{E}[L],$$

$$\mathrm{VaR}_\alpha = \inf\{x : P(L > x) \leq 1 - \alpha\},$$

$$\mathrm{ES}_\alpha = \mathbb{E}[L \mid L \geq \mathrm{VaR}_\alpha].$$

In credit portfolios, ES is often preferred because it is coherent and focuses on the tail beyond the quantile.

9.3 Stress Design

Stress testing in this framework can be applied at several layers: (i) hazard scaling (increase λ_i by a stress multiplier or replace with TTC hazards), (ii) recovery stress (lower R), (iii) correlation stress (increase systemic correlation or impose a one-factor structure), and (iv) copula stress (reduce ν under Student-t to increase tail dependence). A disciplined capital process defines a small set of stress narratives, maps each narrative to parameter shocks, and justifies the calibration.

9.4 Allocation and Marginal Contributions

Risk allocation in baskets can be approached via incremental capital: bump one name's hazard curve or apply an idiosyncratic correlation shock and re-evaluate capital metrics. Because the payoff depends on order statistics, marginal contributions may be highly non-linear: a single weak name can contribute disproportionately to low-k risk, while systemic dependence drives high-k risk. Euler allocation for VaR/ES can be approximated by scenario attribution when the loss is a smooth function of parameters; otherwise, bump-and-revalue remains a robust operational choice.

10. Model Risk, Validation, and Governance

10.1 Sources of Model Risk

Key model risk drivers include: (a) copula choice (Gaussian vs t, tail structure), (b) correlation estimation and cleaning, (c) the mapping from market spreads to hazards (bootstrap conventions, discount curve, accrual approximation), (d) recovery assumptions, (e) numerical error from finite Monte Carlo, and (f) the measure transformation from risk-neutral hazards to real-world hazards for capital.

10.2 Validation Tests

- **Unit tests**: survival monotonicity, hazard positivity, bootstrap reproduction of input spreads within tolerance.
- **Sanity checks**: basket spreads monotone decreasing in k; spread limits as correlation approaches 0 and 1.
- **Backtesting** (where feasible): compare implied joint default frequencies vs historical clusters (acknowledging limited data).
- **Sensitivity analysis**: correlation, recovery, hazard scaling, t degrees-of-freedom; ensure outputs move in economically consistent directions.

10.3 Governance and Documentation

For capital use, documentation should include model purpose, assumptions, calibration sources, stress design, known limitations, and controls for numerical stability. Model change should be managed through version control and peer review, with reproducible runs and archived input datasets. Independent validation should review both methodology and implementation (code review, replication, and challenge).

11. Python Implementation Architecture

11.1 Design Principles

The Python refactor decomposes the workflow into modular components: hazard curve representation, CDS bootstrap, copula sampling, default-time inversion, and contract valuation. This separation improves readability, testability, and extensibility. It also supports a risk-capital pipeline where the same scenario engine is reused across valuation and capital runs.

11.2 Key Functions and Data Structures

- `HazardCurve`: piecewise-constant hazard representation with `survival()` and `inverse_default_time()`.
- `bootstrap_hazard_from_cds()`: sequential calibration via root finding to par spreads.
- `sample_copula_uniforms()`: Gaussian and Student-t copula sampling using Cholesky and scale-mixture.
- `price_nth_to_default_mc()`: PV_{prot}, $PV01$, and fair spread.
- `loss_distribution_at_horizon()`: EL, VaR, ES for a conservative horizon loss proxy.

11.3 Reproducibility and Performance

Reproducibility is achieved through explicit random seeds and deterministic inputs. Performance is driven by NumPy vectorisation; for larger problems, one may incorporate just-in-time compilation or GPU acceleration, but the current design aims to be readable and correct first. For capital reporting, deterministic builds and pinned dependencies are recommended.

12. Extensions and Research Agenda

Several extensions naturally follow. First, dynamic spread modelling can be integrated by simulating hazard rate processes or spread processes and computing mark-to-market at horizon. Second, stochastic recovery and recovery-default correlation can be introduced. Third, a factor intensity model can replace the copula, yielding a structural dependence through shared systematic factors. Fourth, one can incorporate wrong-way risk via coupling between discount factors and default intensities. Finally, robust calibration and uncertainty quantification can be formalised via Bayesian approaches or bootstrapped confidence intervals for correlation and hazard estimates.

In a capital environment, the key objective is not maximum realism but a defensible mapping from stress narratives to quantitative outputs. The model should therefore emphasise transparency, sensitivity analysis, and a clear control framework.

Appendix A. Derivation of CDS Par Spread under Discretisation

This appendix summarises the discrete approximation used for CDS valuation. Under deterministic discounting and survival $S(t)$, the protection PV is approximated by summing discounted default probabilities over coupon intervals. Let $t_0 = 0$ and t_j be coupon dates. The default probability in $(t_{j-1}, t_j]$ is approximately $S(t_{j-1}) - S(t_j)$. The expected discounted protection payment is

$$PV_{\text{prot}} = (1 - R) \sum_j D\, F(t_j)\big(S(t_{j-1}) - S(t_j)\big).$$

The premium PV per unit spread is

$$PV01 = \sum_j D\, F(t_j)\, \delta_j\, S(t_j) + \sum_j D\, F(t_j) \cdot 1/2\, \delta_j\big(S(t_{j-1}) - S(t_j)\big),$$

where the second term is the half-period accrual approximation for analytic settings. In simulation, we compute accrual exactly pathwise.

The par spread s^* satisfies $PV_{\text{prot}} = s^* \cdot PV01$. Bootstrapping chooses hazards such that s^* matches market quotes at each maturity.

Appendix B. Inverting a Piecewise-Constant Cumulative Hazard

Given piecewise-constant hazards λ_k on intervals $(T_{k-1}, T_k]$, the cumulative hazard is

$$H(t) = \sum_{j<k} \lambda_j \, (T_j - T_{j-1}) + \lambda_k (t - T_{k-1}), \quad t \in (T_{k-1}, T_k].$$

To sample default times, draw $U \sim \mathrm{Uniform}(0,1)$ and set $X = -\log U \sim \mathrm{Exp}(1)$. Then solve for t such that $H(t) = X$. Because H is piecewise linear and strictly increasing, inversion is done by locating the first segment where cumulative hazard exceeds X and solving within that segment analytically.

This transformation ensures that τ has the desired survival $S(t) = e^{-H(t)}$ and makes simulation efficient: no numerical root finding per path is required.

Appendix C. Copula Calibration Considerations

In practice, copula parameters can be calibrated using historical data on asset returns or spread changes (pseudo-observations). For the Gaussian copula, one estimates a correlation matrix, often applying shrinkage to improve stability. For the Student-t copula, degrees of freedom ν controls tail dependence; ν can be estimated by maximum likelihood on pseudo-observations or selected to match observed tail co-movement. In many capital applications, ν is treated as a stress knob rather than a point estimate, reflecting uncertainty and the desire to produce conservative tails.

A critical governance point is that the mapping between observed correlations and default-time dependence is not unique. Therefore, model documentation must describe the proxy choice and provide sensitivity analysis to alternative proxies and stress levels.

Appendix D. Implementation Excerpt (Conceptual)

The Python module provides a clear separation of responsibilities. `HazardCurve` encapsulates survival and inverse mapping, while the copula sampler creates correlated uniforms. Pricing then reduces to computing $\tau_{(k)}$ and discounting cashflows. For readability in the report we refer users to the repository for full source code, tests, and example scripts.

Appendix E. Glossary

- **CDS**: Credit Default Swap.
- **Basket CDS**: CDS referencing multiple names; payoff depends on a portfolio default event.
- k**th-to-default**: triggers on the k-th default time $\tau_{(k)}$ among names.
- **Hazard rate / intensity** $\lambda_i(t)$: instantaneous default arrival rate under reduced-form modelling.
- **Copula**: function $C: [0,1]^d \to [0,1]$ coupling marginals into a joint distribution (Sklar, 1959).
- **EL, VaR, ES**: expected loss, value-at-risk, expected shortfall.

13. Literature Context and Conceptual Positioning

The modelling of correlated default events in credit portfolios has a long and pragmatic history, partly shaped by data scarcity. Unlike equity returns, defaults are rare, and multi-name joint default observations are rarer still. As a result, multi-name credit derivative models have typically prioritised tractability, calibration to traded instruments, and interpretability under stress. The copula framework is emblematic: it allows practitioners to preserve accurate marginal calibration to single-name CDS while imposing a dependence structure via a small number of parameters — often a correlation matrix Σ and (in heavier-tailed variants) a degrees-of-freedom parameter ν. This separation is operationally attractive, but it also creates an explicit layer where model risk concentrates.

The Gaussian copula gained prominence due to its analytic convenience and the ability to connect dependence to linear correlation proxies. However, post-crisis experience reinforced that a single correlation parameter (or even a full matrix Σ) does not fully describe tail co-movement. For portfolio credit products, the relevant co-movement is not merely contemporaneous correlation in market observables, but clustering in default arrival times, correlated deterioration of credit quality, and common exposure to macroeconomic regimes. The Student-t copula is one of the simplest ways to introduce tail dependence while staying within an elliptical family. In risk capital, this additional flexibility is useful both as a stress lever and as an uncertainty band around dependence.

Parallel to copula approaches, factor intensity models provide a different dependence mechanism: conditional on common systematic factors, names default independently with intensities that are functions of the factors. This can yield more structural interpretability, at the expense of additional calibration complexity. In many capital frameworks, factor models are preferred when a coherent macro linkage is required. Nonetheless, for instruments like NTD baskets, copula models remain valuable as a transparent benchmark and as a rapid stress-testing engine, especially when the objective is to explore sensitivity rather than to claim a definitive structural truth.

13.1 Why Nth-to-Default Remains Instructive

Although market liquidity in bespoke NTD baskets can be episodic, the contract remains a pedagogically and operationally useful object. First, the order-statistic payoff cleanly separates idiosyncratic and systemic drivers: low-k baskets emphasise the weakest name and near-term hazards, while high-k baskets emphasise the probability of multi-name clustering. Second, NTD pricing outputs offer diagnostic signals about dependence assumptions: if two models match single-name spreads but produce materially different high-k spreads, the divergence can be directly interpreted as a difference in tail dependence or systemic clustering. Third, in a capital setting, NTD-style triggers approximate stress loss mechanisms that depend on multiple failures, which is often the relevant case for concentrated sector books.

13.2 A Practical Taxonomy of Dependence Choices

From a risk management perspective, dependence modelling can be organised into four broad categories: (i) static copulas on default times (as used here), (ii) dynamic copulas on latent credit factors, (iii) factor intensity models with systematic drivers, and (iv) structural models based on firm value processes. The choice is driven by the use case. For pricing and rapid sensitivity analysis, static copulas are attractive. For long-horizon capital or stress testing, factor intensity models provide a more explicit macro channel. For deep research into joint credit-equity dynamics, structural models can be useful, but are typically too parameter-heavy for routine capital production. In practice, organisations often maintain a hierarchy: a baseline copula engine for diagnostics and benchmarking, and a factor-based capital engine for regulatory narratives.

14. Mathematical Deep Dive: Order Statistics and Loss Mechanics

14.1 Distribution of the k-th Default Time

Let $\tau_1, \dots, \tau_d$ denote default times with joint distribution implied by a copula C and marginals F_i. The k-th order statistic $\tau_{(k)}$ is defined as the k-th smallest element of the set. Its distribution depends on the full joint dependence structure, not merely on pairwise correlations. Even in the independent case, the distribution is a combinatorial object:

$$P(\tau_{(k)} > t) = P(\text{at most } k-1 \text{ defaults by time } t) = \sum_{j=0}^{k-1} P\,(\text{exactly } j \text{ defaults by } t).$$

Under independence, each term factorises over subsets of names; under dependence, no such factorisation exists, and simulation is the natural route.

Nonetheless, it is useful to highlight two limiting dependence regimes. Under perfect comonotonicity (an extreme tail-dependent case), all default times are monotone transforms of a single latent variable, so $\tau_{(1)} = \cdots = \tau_{(d)}$ almost surely. In this regime, all kth-to-default contracts behave similarly. Under independence, $\tau_{(k)}$ spreads out, and higher k triggers become progressively rare. Realistic portfolios lie between these extremes, with dependence creating clustering especially in stressed regimes.

14.2 Protection Leg as a Stopped Payoff

The protection payment for a kth-to-default is a stopped payoff at $\tau_{(k)}$. The random time introduces an optionality-like feature: the payoff is only realised if sufficient defaults occur before maturity. For high k, this resembles a deep out-of-the-money tail option on the portfolio credit process. This perspective is useful for capital: tail options are precisely where dependence assumptions are most consequential and where model uncertainty should be managed explicitly.

14.3 Premium Leg and Accrual Subtleties

The premium leg is a pathwise sum of deterministic coupon amounts multiplied by the survival indicator that the contract is alive. Accrued premium on default introduces a dependence on the exact location of $\tau_{(k)}$ within the coupon interval. While this is a second-order effect for many pricing tasks, it is straightforward to implement exactly in Monte Carlo and avoids unnecessary approximation error. In a capital setting, accuracy in the mapping from scenario to cashflow is desirable because tail quantiles can be sensitive to small systematic biases.

14.4 Rare-Event Behaviour for Large k

As k increases, the default trigger can become a rare event, particularly for portfolios of higher credit quality. Rare-event simulation raises two operational issues: (i) naive Monte Carlo may require a very large number of paths to obtain stable estimates of PV and, even more so, tail risk measures; (ii) variance reduction becomes critical. In this note we keep the engine transparent and rely on large path counts for stability. In production, importance sampling or conditional Monte Carlo can materially reduce computational cost.

14.5 Importance Sampling Sketch (Optional Production Extension)

A common importance sampling idea for portfolio default simulation is to shift the systematic factor distribution so that default clustering is more likely, then re-weight paths by likelihood ratios. For Gaussian copula with a one-factor structure, one can tilt the factor to increase joint default probability. For Student-t copulas, one can condition on low realisations of the mixing variable V. The benefit is a dramatic reduction in variance for high-k protection PV and for tail quantiles. The cost is additional complexity and the need for careful validation of weights to avoid numerical instability.

15. Risk-Capital Integration: From Instrument to Portfolio Capital

15.1 Where the Basket Sits in a Capital Stack

In a risk-capital environment, the basket CDS is rarely evaluated in isolation. It is embedded in a portfolio of credit exposures that may include single-name CDS, index CDS, cash bonds, loans, and structured credit. The basket can be interpreted as a non-linear exposure to joint default clustering, and thus as an amplifier (or hedge) of portfolio tail risk. A capital model must therefore support aggregation and allocation, which motivates a consistent scenario generation mechanism across exposures.

15.2 Capital Horizon Choices and Measure Alignment

Capital horizons vary by framework: internal economic capital may use 1 year at 99.9%, market risk frameworks may use 10 days at 99% (with scaling), and stress testing may consider multi-quarter horizons. Reduced-form default-time simulation is naturally aligned to horizons where defaults are plausible drivers (months to years). If a short horizon is required, one typically models spread moves and jump-to-default exposure rather than realised defaults. Accordingly, for a 1-year capital horizon we can define loss as:

$$L = \mathrm{LGD} \cdot \frac{N}{d} \cdot DF(\tau_{(k)}) \cdot \mathbf{1}\{\tau_{(k)} \leq 1\},$$

while for shorter horizons the relevant loss is the MtM change of the instrument conditional on spread dynamics.

15.3 Stress Scenarios: Systematic and Idiosyncratic Components

A robust stress design decomposes parameter shocks into systematic and idiosyncratic components. Systematic stress increases joint distress via higher systemic correlation, lower ν (higher tail dependence), and hazard scaling tied to macro downturn. Idiosyncratic stress targets specific names (e.g., a weak bank in the basket) by shifting its hazard curve upward or lowering its recovery. This decomposition enables a transparent mapping from narrative to numbers, which is essential for governance.

- **Systemic downturn**: multiply all hazards by a factor (e.g., $2\times$), increase average correlation (e.g., $+0.15$), set Student-t $\nu = 4$.
- **Idiosyncratic shock**: bump a selected name's hazard curve by $+50\%$ and reduce recovery by 10 points; keep other names unchanged.
- **Concentration stress**: replace Σ by a one-factor matrix with higher common loading to represent crowded exposures.

15.4 Capital Metrics Beyond VaR/ES

Beyond EL, VaR, and ES, capital frameworks may require incremental metrics: stressed loss under specified scenarios, diversification benefit (standalone vs portfolio capital), and capital attribution by driver (hazard vs correlation). For NTD baskets, driver attribution is particularly useful: it clarifies whether capital is driven by a weak-name idiosyncratic risk or by systemic clustering. Scenario attribution can be performed by recording, for each tail scenario, which names defaulted and in what order, and by tracking the distribution of $\tau_{(k)}$ conditional on the tail event.

15.5 Backtesting and Benchmarking in Sparse Data

Backtesting default clustering is challenging due to sparse observations. Practical validation therefore uses multiple complementary benchmarks: (i) compare implied joint default probabilities under the model with those implied by index tranche markets (when available), (ii) compare dependence proxies across markets (equity, CDS spreads, macro factors), and (iii) stress reasonableness: ensure that under a severe macro stress the model produces loss magnitudes consistent with historical crisis episodes. Where direct backtests are infeasible, documentation should emphasise the benchmarking logic and the conservatism of stress settings.

16. Implementation Engineering: Testing, Packaging, and Reproducibility

16.1 Unit Tests and Numerical Tolerances

A production-grade implementation should include unit tests for each layer. For hazard curves, tests should verify survival monotonicity and ensure that bootstrapped par spreads reproduce input spreads within a defined tolerance (e.g., 0.1 bps). For copulas, tests should verify that simulated uniforms are marginally uniform and that the implied rank correlation is consistent with the target Σ in large samples. For pricing, tests should cover limiting cases such as $k = 1$ and $k = d$, and correlation approaching 0 and 1, to detect implementation errors.

16.2 Packaging and Dependency Control

For reproducibility, pin dependencies (NumPy, SciPy) and record the random seed, input spread set, and correlation matrix used for each run. In regulated environments, artefact capture is important: store the exact code version (git commit hash), inputs, and outputs. If the model is used for capital reporting, adopt a release process with peer review and change control.

16.3 Documentation and MkDocs Site Integration

A lightweight documentation site improves auditability. MkDocs can host the model description, calibration notes, and reproducibility instructions. For example, include pages for model overview, calibration, risk-capital outputs, and governance controls. Publish the docs using GitHub Pages so that stakeholders can access a stable, versioned reference. This also encourages disciplined updates: every model change should be reflected in documentation.

16.4 Suggested Repository Structure

1. `src/` for the Python module (or keep a single file for a small project).
2. `docs/` for MkDocs markdown pages.
3. `data/` (optional) for sample inputs with clear licensing and anonymisation.
4. `tests/` for unit tests (pytest).
5. `notebooks/` (optional) for exploratory analysis (export key results to docs).

Appendix F. Parameter Uncertainty and Conservative Overlays

Capital models must address parameter uncertainty explicitly. For this framework, key uncertain parameters include hazard curves (especially beyond the last liquid CDS tenor), recovery, correlation matrix estimation error, and Student-t degrees of freedom ν. Conservative overlays can be applied by adopting upper confidence bounds for hazards, lower bounds for recovery, and stressed correlations. A practical approach is to define a baseline and a small set of conservative overlays, then to report a range of capital outputs. This aligns with model risk management: it communicates that the capital number is not a single point estimate but a controlled range under plausible parameter perturbations.

Appendix G. Worked Example Narrative for Capital Committees

When presenting results to a capital committee, the narrative should emphasise: (i) the instrument's non-linearity and sensitivity to clustering, (ii) the calibration anchor to market spreads, (iii) the stress levers and why they are conservative, and (iv) how outputs change under stress. A succinct committee package includes a baseline fair spread table across k, a stress table showing EL/VaR/ES at the capital horizon, and a sensitivity chart showing the impact of correlation and ν. While this report does not embed figures, the accompanying repository can generate them for documentation.

Appendix H. Reproducibility Checklist

- Record input spreads and their date/time source.
- Record discount curve assumptions and recovery R.
- Record dependence inputs: Σ, proxy used, and any cleaning/shrinkage method.
- Record simulation settings: copula choice, ν (if applicable), number of paths N, seed.
- Archive outputs with commit hash and runtime environment (Python version, package versions).

17. Correlation Modelling in Practice

17.1 Estimation from Market Observables

The dependence input Σ is frequently the most debated ingredient in a copula-based credit model. Unlike single-name hazards, which can be bootstrapped from liquid CDS spreads, default dependence is not directly observable for most bespoke portfolios. Practitioners therefore rely on proxies. Three common choices are: (i) equity return correlations, (ii) CDS spread change correlations, and (iii) correlations implied by index tranches or bespoke tranche quotes. Each proxy has strengths and weaknesses. Equity correlations are abundant and stable but may overstate short-horizon co-movement relative to default-time dependence. CDS spread correlations are closer to credit risk but are noisier and can be regime-dependent. Tranche-implied correlations are theoretically closest to default clustering but require liquid tranche markets and a consistent mapping between tranche and basket dependence.

In risk capital settings, the objective is typically not to claim a unique "true" Σ, but to define a defensible baseline and a set of conservative stresses. The baseline may use a long historical window with shrinkage to reduce noise. Stresses may include correlation uplift, increased systematic factor loading, or replacement of Σ by a conservative one-factor structure. The model should also be robust to estimation error; for example, a small perturbation to Σ should not cause discontinuous changes in basket spreads unless the portfolio is near a cliff (e.g., dominated by a single weak name).

17.2 Cleaning, Shrinkage, and Positive Semidefiniteness

Empirical correlation matrices computed from finite samples can be ill-conditioned or not positive semidefinite. This is not merely a numerical nuisance; it is also a signal of estimation noise. Common remedies include:

- **Diagonal loading**: add a small $\varepsilon > 0$ to the diagonal to ensure Cholesky factorisation succeeds.
- **Eigenvalue clipping**: replace negative eigenvalues with a small positive floor, then renormalise to unit diagonal.
- **Shrinkage**: blend the sample correlation with a structured target (identity, constant-correlation, or factor model).
- **Factor modelling**: estimate a low-rank factor covariance and add idiosyncratic variance on the diagonal.

For risk capital, shrinkage has an interpretation aligned with conservatism and robustness: it reduces overfitting to idiosyncratic noise and stabilises capital metrics across time. The chosen cleaning method should be documented, and sensitivity to alternative cleaning should be demonstrated at least at model validation time.

17.3 One-Factor and Multi-Factor Structures

A useful structured alternative to a full empirical Σ is a one-factor correlation model:

$$Z_i = a_i\, M + \sqrt{1 - a_i^2}\, \varepsilon_i,$$

where $M \sim \mathcal{N}(0,1)$ is a common factor and $\varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$ are independent idiosyncratic normals, with $M \perp \varepsilon_i$ for all i. This implies

$$\mathrm{Corr}(Z_i, Z_j) = a_i a_j, \quad i \neq j.$$

The one-factor model is appealing for capital because it makes systematic clustering explicit and often produces more conservative high-k risk than a noisy full matrix. Multi-factor extensions allow sector and region factors and are a natural bridge to factor intensity models.

17.4 Regime Dependence and Stress Calibration

Correlation is not constant across regimes. In benign markets, idiosyncratic dispersion dominates; in crises, common factors dominate and correlations rise. A practical capital workflow therefore calibrates a baseline Σ_{base} from a long sample and defines a crisis Σ_{stress} from a stressed window (e.g., 2008–2009 or 2020Q1 depending on the portfolio). Alternatively, one can parametrise Σ as a blend:

$$\Sigma(\alpha) = (1 - \alpha)\, \Sigma_{\text{base}} + \alpha\, \Sigma_{\text{stress}}, \quad \alpha \in [0,1],$$

and report capital across α. This produces a smooth stress path and supports governance discussions about stress severity.

18. Calibration Beyond Single-Name Spreads

18.1 Risk-Neutral vs Real-World Transformation

The hazard curves bootstrapped from CDS spreads are risk-neutral objects: they incorporate market prices of default risk and liquidity premia. Risk capital frameworks often require a real-world default view, particularly for multi-year horizons. A common operational transformation is to apply a risk premium adjustment:

$$\lambda^P(t) = \kappa\,\lambda^Q(t), \quad \kappa < 1,$$

where $\kappa < 1$ reduces the implied default intensity toward historical levels (risk-neutral hazards typically exceed real-world hazards due to the credit risk premium), or to calibrate λ^P directly to historical PDs and use λ^Q only for valuation. The choice should be consistent with the capital framework and the intended use of the model (pricing vs capital).

18.2 Benchmarking to Index Tranche Markets

When liquid index tranche quotes exist (e.g., CDX or iTraxx tranches), they provide a valuable dependence benchmark. Although NTD baskets are not tranches, both depend on joint default dynamics. A validation approach is to calibrate a dependence parameter (such as average correlation or factor loading) to match tranche spreads, then reuse that dependence setting for NTD pricing on a similar sector portfolio. This is not perfect, because tranche payoffs depend on portfolio loss fraction rather than order statistics, but it provides an external market anchor for dependence, especially for tail behaviour.

18.3 Heterogeneity and Mapping Challenges

Bespoke baskets are heterogeneous: names differ in hazard term structure, recovery, and sector exposure. This heterogeneity complicates dependence calibration. For example, a correlation estimated from equity returns may reflect high-frequency co-movement that is only loosely connected to default clustering, while tranche-implied correlations are averaged across a broad index and may not transfer cleanly to a concentrated basket. A disciplined approach therefore emphasises ranges and conservative overlays rather than point calibration.

18.4 A Pragmatic Calibration Ladder

A pragmatic calibration ladder for capital practitioners can be summarised as follows:

1. Calibrate single-name hazard curves λ^Q to market spreads for valuation.
2. Select a dependence proxy for Σ (equity, CDS spread, or factor model) and apply shrinkage/cleaning.
3. Select Student-t ν as a baseline (moderate tail) and a stressed value (strong tail), justified by crisis co-movement diagnostics.

4. For capital, define λ^P via historical PD calibration or a risk premium scaling of λ^Q.
5. Validate dependence against available tranche or portfolio benchmarks where possible; otherwise, validate via stress reasonableness and sensitivity analysis.

Appendix I. Capital Case Study Template (Plug-and-Play)

This appendix provides a template for turning model outputs into a capital case study suitable for internal governance. Step 1 (Baseline valuation): report bootstrapped hazards and fair basket spreads across k at maturity T, including Monte Carlo standard errors. Step 2 (Horizon losses): define a capital horizon H (e.g., 1 year) and compute the discounted protection loss distribution for a protection seller. Report EL, $\mathrm{VaR}_{99\%}$, $\mathrm{VaR}_{99.5\%}$, $\mathrm{ES}_{99\%}$, $\mathrm{ES}_{99.5\%}$. Step 3 (Stresses): define 3–5 stresses and repeat Step 2. Step 4 (Attribution): for tail scenarios (e.g., top 1% losses), record the frequency of each name appearing among the first k defaults and the distribution of $\tau_{(k)}$.

- **Stress A (Systemic)**: hazards $\times\, 2$, correlation $+0.15$, $\nu = 4$, recovery -10 pts.
- **Stress B (Idiosyncratic)**: weakest name hazards $\times\, 3$, recovery -15 pts, others unchanged.
- **Stress C (Correlation shock)**: replace Σ with one-factor structure, increase common loading by $+0.10$.
- **Stress D (Tail shock)**: keep Σ fixed, reduce ν from 8 to 3 to amplify tail dependence.

The committee narrative should emphasise interpretability: explain why each stress is plausible, how it relates to historical episodes, and why outputs are conservative. Where quantitative calibration is weak, acknowledge uncertainty and provide a range rather than a single number.

Appendix J. Suggested Figures and Tables for a 20-Page Report

To make the report visually and analytically complete, the following figures and tables are recommended. They can be generated from the Python example script and embedded into documentation:

- Figure 1: Bootstrapped survival curves $S_i(t)$ for each name (1–5Y).
- Figure 2: Histogram of $\tau_{(k)}$ for $k = 1$ and $k = d$ under Gaussian vs Student-t.
- Figure 3: Basket fair spread vs k for multiple correlation levels (ρ grid).
- Figure 4: Loss distribution at horizon H under baseline and stressed settings.
- Table 1: Input CDS spreads by tenor (already included).
- Table 2: Bootstrapped hazard rates by segment for each name.
- Table 3: Fair spreads across k under Gaussian and Student-t (already included).
- Table 4: Capital metrics (EL, VaR, ES) under baseline and stresses.

Appendix K. Detailed Algorithmic Specification

K.1 Inputs

The model inputs are grouped into four categories. Market calibration inputs: (i) single-name CDS par spreads across maturities, (ii) discount curve assumptions (flat rate r or full curve), (iii) recovery R. Dependence inputs: (i) correlation matrix Σ, (ii) copula choice (Gaussian or Student-t with ν). Contract inputs: (i) maturity T, (ii) trigger threshold k, (iii) notional N and notional allocation convention, (iv) premium payment frequency. Simulation inputs: number of Monte Carlo paths N_{paths}, random seed, and any variance reduction toggles.

K.2 Outputs

Primary valuation outputs: PV_{prot}, $PV01$ (including accrual), and fair spread s^* in bps. Risk-capital outputs: EL, VaR_α, ES_α at selected confidence levels, and stress tables. Diagnostic outputs: Monte Carlo standard errors and scenario attribution summaries.

K.3 Step-by-Step Procedure (Valuation)

1. For each name i, bootstrap a piecewise-constant hazard curve from its CDS par spreads: solve sequentially for segment hazards so that model par spreads match market quotes.
2. Construct or import the copula dependence matrix Σ, applying cleaning/shrinkage if required to ensure positive semidefiniteness.
3. Simulate latent vectors $\boldsymbol{Z}_n \sim \mathcal{N}(\boldsymbol{0}, \Sigma)$ (Gaussian) or scale-mixture t vectors (Student-t). Convert to uniforms $\boldsymbol{U}_n$ via the appropriate CDF mapping.
4. For each simulated path n and each name i, compute $\tau_i^{(n)} = H_i^{-1}(-\log U_i^{(n)})$.
5. For each path n, compute $\tau_{(k)}^{(n)}$ as the k-th order statistic of $\tau_1^{(n)}, \dots, \tau_d^{(n)}$.
6. Compute pathwise protection payoff: $\text{LGD} \cdot (N/d) \cdot \mathbf{1}\{\tau_{(k)}^{(n)} \le T\} \cdot DF(\tau_{(k)}^{(n)})$.
7. Compute pathwise premium $PV01$: sum discounted coupon year-fractions for coupon dates up to $\min(\tau_{(k)}^{(n)}, T)$, plus accrued premium from last coupon to $\tau_{(k)}^{(n)}$ if default occurs before T.
8. Average across paths to obtain PV_{prot} and $PV01$. Compute $s^* = PV_{\text{prot}}/PV01$ and report in bps.

K.4 Step-by-Step Procedure (Risk Capital)

A capital extension reuses the scenario generator but replaces the valuation objective with a loss distribution objective. The simplest conservative loss definition for the protection seller is the discounted protection payment if $\tau_{(k)}$ occurs within a horizon H.

1. Select capital horizon H (e.g., 1 year) and define loss $L = \mathrm{LGD} \cdot (N/d) \cdot DF(\tau_{(k)}) \cdot \mathbf{1}\{\tau_{(k)} \leq H\}$.
2. Run Monte Carlo under the chosen parameterisation (baseline and stresses).
3. Compute $EL = \bar{L}$, $\mathrm{VaR}_\alpha = \hat{Q}_\alpha(L)$, and $\mathrm{ES}_\alpha = \mathbb{E}[L \mid L \geq \mathrm{VaR}_\alpha]$ for $\alpha \in \{99\%, 99.5\%\}$.
4. For tail attribution, subset paths with $L \geq \mathrm{VaR}_{99\%}$ and tabulate: default order frequencies, distribution of $\tau_{(k)}$, and contribution of each name to being among early defaults.
5. Report a stress table and explain the mapping from stress narratives to parameter shocks.

Appendix L. Capital Allocation and Sensitivity Math (Practitioner Note)

L.1 Incremental Risk and Bump-and-Revalue

Because the kth-to-default payoff depends on an order statistic, it is not generally differentiable with respect to all parameters in a smooth way. This complicates pure analytic Euler allocations. Nonetheless, bump-and-revalue provides a robust operational allocation mechanism. Define a baseline parameter set $\boldsymbol{\theta}$ and compute a capital metric $C(\boldsymbol{\theta})$ (e.g., $\mathrm{ES}_{99.5\%}$ of horizon loss). For name i, define a bump to its hazard curve or an idiosyncratic correlation loading, producing $\boldsymbol{\theta}^{(i)}$. The marginal contribution is approximated as

$$\Delta_i = C\left(\boldsymbol{\theta}^{(i)}\right) - C(\boldsymbol{\theta}).$$

This incremental capital can be normalised to allocate total capital across names or to attribute capital between idiosyncratic and systemic drivers.

L.2 Scenario Attribution for Tail Sets

A complementary allocation method uses scenario attribution. Let $\mathcal{T}_\alpha$ be the set of paths whose loss exceeds VaR_α. Within $\mathcal{T}_\alpha$, compute empirical frequencies of each name appearing among the first k defaults, and compute conditional averages of each name's default time. This produces an interpretable narrative: in the tail, which names tend to fail early, and how quickly does clustering occur. Because high-k baskets are systemic by construction, tail sets often show concentrated systemic patterns rather than single-name dominance.

L.3 Sensitivities to Dependence Parameters

Sensitivities to dependence parameters can be summarised using stress ladders. For example, create a grid of average correlations $\rho \in [0.0, 0.8]$ and degrees-of-freedom $\nu \in \{3, 4, 6, 10, \infty\}$. For each grid point, compute fair spreads and capital metrics. Plot surfaces or heatmaps (in the documentation site) to show how capital changes as dependence strengthens.

L.4 Importance Sampling for Tail Capital (Advanced)

Tail capital metrics, especially ES at high confidence, can be expensive to estimate with naive Monte Carlo. Consider a one-factor Gaussian copula where $Z_i = a_i M + \sqrt{1 - a_i^2}\,\varepsilon_i$. Tail losses correspond to low M. An importance sampling scheme shifts the distribution of M by a tilt parameter μ and reweights paths using the likelihood ratio:

$$\frac{f_M(m)}{f_M(m-\mu)} = \exp(-\mu m + 1/2\,\mu^2).$$

The simulation visits tail regions more frequently, dramatically reducing variance for high-k protection PV. A practical implementation requires: (i) choosing μ to balance variance and stability, (ii) monitoring weight degeneracy (effective sample size), and (iii) validating reweighted estimates against brute-force Monte Carlo for test cases. For Student-t copulas, similar ideas apply by conditioning on low realisations of the mixing variable V.

L.5 Governance Note on Advanced Variance Reduction

While importance sampling can materially improve efficiency, it increases implementation complexity and model risk. For capital production, a common governance posture is to keep the baseline engine simple and transparent (large path counts, straightforward sampling) and to use advanced variance reduction as an optional acceleration tool subject to additional validation controls. Documentation should separate methodology (what is being estimated) from numerics (how it is estimated).

Appendix M. Reporting Template for Capital Governance

M.1 Model Definition (One Page)

- Instrument: kth-to-default basket CDS, maturity T, payment frequency, notional convention, recovery R.
- Marginals: reduced-form intensity model, piecewise-constant hazards calibrated to CDS spreads (valuation view: λ^Q).
- Dependence: copula choice and parameterisation (Σ, and ν if Student-t).
- Numerics: Monte Carlo paths N_{paths}, variance reduction settings, seed policy, convergence diagnostics.

M.2 Inputs (One Page)

Inputs should be presented in a compact table with provenance. For market inputs, document the data date, source, and any cleaning. For correlation inputs, document the proxy (equity returns, CDS spread changes), sample window, and shrinkage method. For capital inputs, document any transformation from λ^Q to λ^P, including risk premium assumptions or historical PD sources.

M.3 Outputs (One Page)

Outputs should include valuation metrics and capital metrics. Valuation metrics: PV_{prot}, $PV01$, and fair spread s^* in bps. Capital metrics: EL, VaR_α and ES_α at prescribed confidence levels, and stress losses under defined stress narratives. For each metric, include numerical uncertainty measures (standard errors, and where feasible, tail uncertainty via batch resampling).

M.4 Sensitivities and Model Risk Summary (One Page)

Sensitivities should focus on the key drivers: correlation uplift $\Delta\rho$, tail dependence ν, hazard scaling, and recovery stress. For each driver, present a small ladder showing how capital changes. Conclude with a model risk summary identifying the assumptions with highest impact and specifying conservative overlays (e.g., "use Student-t $\nu = 4$ for stressed capital" or "apply $+15\%$ correlation uplift as a dependency uncertainty add-on").

Appendix N. Code Walkthrough for Practitioners

The accompanying Python refactor is designed to be readable. A quick walkthrough helps practitioners extend the model responsibly. The code centres on the `HazardCurve` object, which stores segment endpoints and hazard rates and provides two critical methods: `survival(t)` and `inverse_default_time(u)`. The `inverse_default_time` method is the workhorse for simulation: it converts copula uniforms U into default times via $\tau = H^{-1}(-\log U)$ without per-path root finding.

Calibration is handled by `bootstrap_hazard_from_cds()`. This routine iterates over maturities, each time solving for the segment hazard that makes the model par spread match the quoted spread. The solver uses robust bracketing and a one-dimensional root finder. In production, you may wish to plug in a discount curve object or to include day count conventions; the structure is modular enough to do so.

Dependence is handled by `sample_copula_uniforms()`. For Gaussian copula, it draws standard normals, applies a Cholesky factor of Σ, and maps to uniforms via Φ. For Student-t, it uses a scale-mixture construction (drawing $W \sim \chi^2(\nu)$ and dividing by $\sqrt{W/\nu}$) that introduces a common random scale per path. The function `_chol_psd()` includes a defensive approach to near-PSD matrices via jitter and eigenvalue cleaning.

Pricing is handled by `price_nth_to_default_mc()`. It simulates default times, extracts $\tau_{(k)}$, and computes discounted protection and premium $PV01$ with accrued premium. The output includes fair spread s^* in bps and leg PVs. For risk capital, `loss_distribution_at_horizon()` uses the same scenario generator but returns $EL/\mathrm{VaR}/\mathrm{ES}$ for a horizon loss proxy.

Appendix O. Illustrative Stress Table and Interpretation Guidance

This appendix provides an illustrative layout for a stress table. Populate the table by running the Python engine under each stress configuration.

Scenario	Hazard scale	Corr uplift	ν	$\mathrm{VaR}_{99.5\%}$	$\mathrm{ES}_{99.5\%}$
Baseline	$1.0\times$	$+0.00$	6	X	X
Systemic downturn	$2.0\times$	$+0.15$	4	X	X
Idiosyncratic shock	$3.0\times$ (weak name)	$+0.00$	6	X	X
Tail stress	$1.5\times$	$+0.10$	3	X	X

Interpretation guidance: hazard scaling moves marginal default probabilities and therefore shifts both EL and tail losses upward, especially for low-k baskets. Correlation uplift and lower ν primarily affect clustering. Their impact is typically most visible for higher k, where multiple defaults are required. If under a systemic downturn stress the incremental effect of correlation uplift is small, it may indicate that hazards already dominate default likelihood; conversely, if correlation uplift materially increases ES for higher k, it indicates that tail clustering is the principal risk driver.

A recommended narrative is to separate *frequency* effects from *severity* effects. Hazard scaling increases frequency of default events. Tail dependence increases the likelihood that several defaults occur close together, which is a severity amplifier for high-k triggers. This decomposition helps committees understand why dependence modelling is not optional for baskets: it is the mechanism by which systemic crises become non-linear losses.

Appendix P. Expanded Glossary and Key Conventions

- **Accrued premium**: premium owed from the last coupon date to the default date when default occurs between coupon dates.
- **Bootstrapping**: sequential calibration method solving one segment parameter at a time to match market quotes.
- **Cumulative hazard** $H(t) = \int_0^t \lambda(s)\, ds$; survival is $S(t) = e^{-H(t)}$.
- **Copula uniform** U_i: random variable on $[0,1]$ that couples marginals; default time obtained by inverse mapping $\tau_i = F_i^{-1}(U_i)$.
- **Degrees of freedom** ν: Student-t parameter controlling tail dependence; lower ν implies heavier tails.
- **Economic capital**: internal capital held against unexpected losses at a high confidence level over a specified horizon.
- **EL / VaR / ES**: expected loss; quantile-based loss threshold; conditional mean beyond the quantile threshold.
- **Order statistic** $\tau_{(k)}$: the k-th smallest value in a sample; here, the k-th default time.
- **PSD matrix**: positive semidefinite matrix required for valid correlations and Cholesky factorisation.
- **Shrinkage**: blending a noisy empirical correlation matrix with a structured target to improve stability.

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Appendix Q. Limitations and Next-Step Roadmap

It is important to state clearly what this framework does not do, particularly when used in a risk-capital context. First, the model is *static* in the sense that it does not simulate spread dynamics or mark-to-market evolution. For capital horizons shorter than the typical default horizon, MtM changes may dominate realised defaults, and a full capital engine should model spread processes and their correlation to the systematic credit factor. Second, recovery is treated as deterministic. In severe stress, recovery uncertainty and recovery-default correlation can meaningfully alter tail losses; incorporating stochastic recovery is a valuable extension. Third, dependence is imposed via an elliptical copula with a fixed correlation matrix. Real-world dependence can be asymmetric and regime-dependent; extensions include regime-switching copulas, time-varying correlations, or factor intensity models that tie dependence to macro states.

A pragmatic roadmap for extending this work in an enterprise capital stack is as follows. (1) Add a discount curve object and day count conventions to align valuation precisely with desk standards. (2) Add calibration and storage of correlation proxies, including shrinkage settings and stressed windows, to ensure auditability. (3) Add a spread-dynamics layer for short-horizon capital, for example by modelling hazard-rate changes as a mean-reverting process with jumps, and compute mark-to-market at horizon conditional on survival. (4) Add a portfolio wrapper that aggregates multiple instruments and supports allocation across desks and legal entities. (5) Add a test suite (pytest) and continuous integration checks to ensure numerical stability and regression control as the model evolves.